University of Zimbabwe

HMTH101 CALCULUS 1

Calculus of Single Variables

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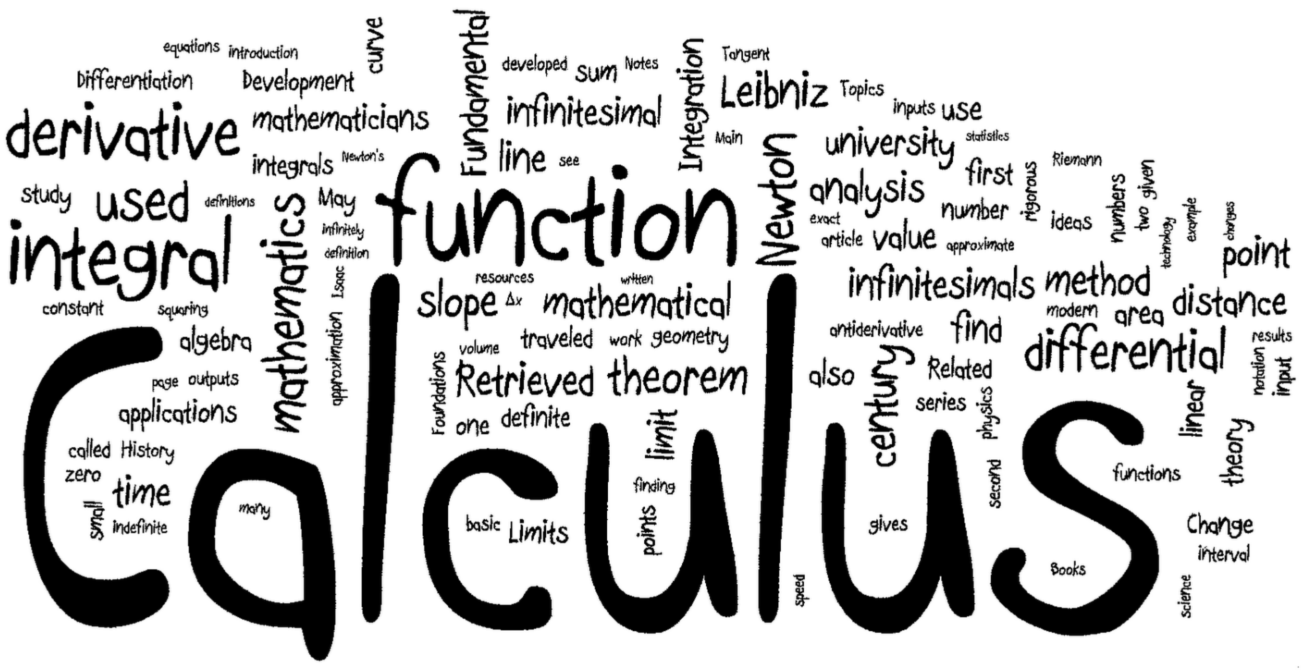

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Contents

Chapter 4

Limits and Continuity

A man has one hundred dollars and you leave him with two dollars. That's subtraction.

—Mae West

The single most important idea in calculus is the idea of limit. More than 2000 years ago, the ancient Greeks wrestled with the limit concept, and they did not succeed. It is only in the past 200 years that we have finally come up with a firm understanding of limits. The study of calculus went through several periods of increased mathematical rigour beginning with the French mathematician Augustin-Loius Cauchy (1789-1857) and later continued by the German mathematician, and former high school teacher, Karl Wilhelm Weierstrass (1815-1897).

4.1 Limit of a Function

If f is a function, then we say $\lim_{x\to a} f(x) = A$, if the value of $f(x)$ gets arbitrarily closer to A as x gets closer and closer to a. For example, $\lim_{x\to 3} x^2 = 9$, since x^2 gets arbitrarily close to 9 as x approaches as close as one wishes to 3.

The definition can be stated more precisely as follows : $\lim_{x\to a} f(x) = A$ if and only if, for any chosen positive number ε, however small, there exists a positive number δ, such that, whenever $0 < |x - a| < \delta$, then $|f(x) - A| < \varepsilon$.

$\lim_{x\to a} f(x) = A$ means that $f(x)$ can be made as close as desired to A by making x close enough, but not equal to a. How close is "close enough to a" depends on how close one wants to make $f(x)$ to A. It also of course depends on which function f is and on which number a is. The positive number ε is how close one wants to make $f(x)$ to A ; one wants the distance to be no more then ε. The positive number δ is how close one will make x to a ; if the distance from x to a is less than δ (but not zero), then the distance from $f(x)$ to A will be less than ε. Thus δ depends on ε. The limit statement means that no matter how small ε is made, δ can be made smaller enough. The letters ε and δ can be understood as "error" and "distance". In these terms the error (ε) can be made as small as desired by reducing the distance (δ).

The $\varepsilon - \delta$ definition of $\lim_{x\to a} f(x) = A$

For any chosen positive number ε, however small, there exists a positive number δ, such that, whenever $0 < |x - a| < \delta$, then $|f(x) - A| < \varepsilon$.

Example: Show that $\lim_{n\to 1}(x^2 + 1) = 2$.

Solution: Need to find δ so that, for a given ε, $|x^2 + 1 - 2| < \varepsilon$ for $|x - 1| < \delta$.

Now

$$\begin{aligned} x^2 + 1 - 2 &= x^2 - 1 \\ &= (x+1)(x-1). \end{aligned}$$

Choose $|x - 1| < 1$ so that $-1 < x - 1 < 1 \Rightarrow 0 < x < 2 \Rightarrow 1 < x + 1 < 3$. You have $|x^2 + 1 - 2| < \varepsilon$ if $3|x - 1| < \varepsilon$ or $|x - 1| < \dfrac{\varepsilon}{3}$. You have now two conditions on x :

$$|x - 1| < 1 \quad \text{and} \quad |x - 1| < \frac{\varepsilon}{3}.$$

Choose $\delta = \min\{1, \frac{\varepsilon}{3}\}$. For a given $\varepsilon > 0$, choose $\delta = \min\{1, \frac{\varepsilon}{3}\}$, then we have $|x - 1| < \delta$, it would be true that $|x^2 + 1 - 2| < \varepsilon$.

Example: Show that $\lim_{x\to 2}(x^2 + 3x) = 10$.

Solution: Let $\varepsilon > 0$. We must produce a $\delta > 0$ such that, whenever $0 < |x - 2| < \delta$ then $|(x^2 + 3x) - 10| < \varepsilon$. First we note that

$$|(x^2 + 3x) - 10| = |(x - 2)^2 + 7(x - 2)| \leq |x - 2|^2 + 7|x - 2|.$$

Also, if $0 < \delta \leq 1$, then $\delta^2 \leq \delta$. Hence, if we take δ to be the minimum of 1 and $\dfrac{\varepsilon}{8}$, then, whenever $0 < |x - 2| < \delta$,

$$|(x^2 + 3x) - 10| < \delta^2 + 7\delta \leq \delta + 7\delta = 8\delta \leq \varepsilon.$$

You Try It: Prove that $\lim_{x\to 0} x \sin\left(\dfrac{1}{x}\right) = 0$.

Right and Left Limits

Considering x and a as points on the real axis where a is fixed and x is moving, then x can approach a from the right or from the left. We indicate these respective approaches by writing $x \to a^+$ and $x \to a^-$.

If $\lim_{x\to a^+} f(x) = A_1$ and $\lim_{x\to a^-} f(x) = A_2$, we call A_1 and A_2 respectively the *right* and *left* hand limits of $f(x)$ at a.

We have $\lim_{x\to a} f(x) = A$ if and only if $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = A$. The existence of the limit from the left does not imply the existence of the limit from the right and conversely. When a function f is defined on only one side of a point a, then $\lim_{x\to a} f(x)$ is identical to the one-sided limit, if it exists. For example, if $f(x) = \sqrt{x}$, then f is only defined to the right of zero. Hence, $\lim_{x\to 0} \sqrt{x} = \lim_{x\to 0^+} \sqrt{x} = 0$. Of course, $\lim_{x\to 0^-} \sqrt{x}$ does not exist, since $\sqrt{x}$ is not defined when $x < 0$. On the other hand, consider the function $g(x) = \sqrt{\dfrac{1}{x}}$, which is defined only for $x > 0$. In this case, $\lim_{x\to 0^+} g(x)$ does not exist and, therefore $\lim_{x\to 0} g(x)$ does not exist.

4.2 Theorems on Limits

1. If $f(x) = c$, a constant, then $\lim_{x\to a} f(x) = c$.
2. If $\lim_{x\to a} f(x) = A$ and $\lim_{x\to a} g(x) = B$, then
   (a) $\lim_{x\to a} kf(x) = kA$, k being any constant.
   (b) $\lim_{x\to a}[f(x) \pm g(x)] = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x) = A \pm B$.
   (c) $\lim_{x\to a} f(x)g(x) = \lim_{x\to a} f(x) \lim_{x\to a} g(x) = AB$.
   (d) $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)} = \dfrac{A}{B}$, provided $B \neq 0$.

Example: If $\lim_{x\to a} f(x)$ exists, prove that it must be unique.

Solution: Must show that if $\lim_{x\to a} f(x) = A_1$ and $\lim_{x\to a} f(x) = A_2$, then $A_1 = A_2$.

By hypothesis, given any $\varepsilon > 0$ we can find $\delta > 0$ such that

$$|f(x) - A_1| < \frac{\varepsilon}{2} \quad \text{when} \quad 0 < |x - a| < \delta$$
$$|f(x) - A_2| < \frac{\varepsilon}{2} \quad \text{when} \quad 0 < |x - a| < \delta.$$

Then

$$|A_1 - A_2| = |A_1 - f(x) + f(x) - A_2| \leq |A_1 - f(x)| + |f(x) - A_2| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

i.e., $|A_1 - A_2|$ is less than any positive number ε (however small) and so must be zero. Thus $A_1 = A_2$.

Example: Given $\lim_{x\to a} f(x) = A$ and $\lim_{x\to a} g(x) = B$. Prove that

$$\lim_{x\to a}[f(x) + g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) = A + B$$

.

Solution: We must show that for any $\varepsilon > 0$, we can find $\delta > 0$ such that $|(f(x)+g(x))-(A+B)| < \varepsilon$ when $0 < |x - a| < \delta$.

By hypothesis, given $\varepsilon > 0$, we can find $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$|f(x) - A| < \frac{\varepsilon}{2} \quad \text{when} \quad 0 < |x - a| < \delta_1$$
$$|g(x) - B| < \frac{\varepsilon}{2} \quad \text{when} \quad 0 < |x - a| < \delta_2.$$

Then

$$|(f(x) + g(x)) - (A + B)| \leq |f(x) - A| + |g(x) - B| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

when $0 < |x - a| < \delta$ where δ is chosen as the smaller of δ_1 and δ_2.

You Try It: Given $\lim_{x\to a} f(x) = A$ and $\lim_{x\to a} g(x) = B$. Prove that

$$\lim_{x\to a} f(x)g(x) = \lim_{x\to a} f(x) \lim_{x\to a} g(x) = AB$$

.

4.3 Special Limits

1. $\lim_{x\to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x\to 0} \frac{1 - \cos x}{x} = 0.$
2. $\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^x = e, \qquad \lim_{x\to 0^+} (1 + x)^{\frac{1}{x}} = e.$
3. $\lim_{x\to 0} \frac{e^x - 1}{x} = 1, \qquad \lim_{x\to 1} \frac{x - 1}{\ln x} = 1.$

4.4 Methods of Calculating $\lim_{x \to a} f(x)$

If $f(a)$ is defined

If $x = a$ is in the domain of $f(x)$ and a is not an endpoint of the domain, and $f(x)$ is defined by a single expression, then

$$\lim_{x \to a} f(x) = f(a).$$

Example: Find $\lim_{x \to 1}(x + 3)$.

Solution: $\lim_{x \to 1}(x + 3) = 1 + 3 = 4$.

Example: Find $\lim_{x \to 1} \frac{1}{x + 2}$.

Solution: $\lim_{x \to 1} \frac{1}{x + 2} = \frac{1}{1 + 2} = \frac{1}{3}$.

Example: Find $\lim_{x \to 8}(x^2 - 7x + 5)$.

Solution: $\lim_{x \to 8}(x^2 - 7x + 5) = 8^2 - 7(8) + 5 = 13$.

Example: Find $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$.

Solution: $\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = \lim_{x \to 2} \frac{(x + 2)(x - 2)}{x - 2} = \lim_{x \to 2}(x + 2) = 4$.

Functions Defined By More Than One Expression

Suppose that $f(x)$ is defined by one expression for $x < a$ and by a different expression for $x > a$.

Example: Show that $\lim_{x \to 0} \frac{|x|}{x}$ does not exist.

Solution: Notice that

$$\frac{|x|}{x} = \begin{cases} \frac{x}{x} = 1, & \text{if} \quad x \geq 0 \\ -\frac{x}{x} = -1, & \text{if} \quad x < 0. \end{cases}$$

i.e., you seek a limit at $x = 0$ of a function that is defined differently on either side of $x = 0$.

$$\begin{aligned}\lim_{x\to 0^-} \frac{|x|}{x} &= \lim_{x\to 0^-}(-1) = -1.\\ \lim_{x\to 0^+} \frac{|x|}{x} &= \lim_{x\to 0^+}(1) = 1.\end{aligned}$$

Since $\displaystyle\lim_{x\to 0^+} \frac{|x|}{x} \neq \lim_{x\to 0^-} \frac{|x|}{x}$, then $\displaystyle\lim_{x\to 0} \frac{|x|}{x}$ does not exist.

Example: Find $\displaystyle\lim_{x\to 0} \frac{\sin 3x}{x}$.

Solution: Since $\displaystyle\frac{\sin 3x}{x} = 3\left(\frac{\sin 3x}{3x}\right)$. Then

$$\lim_{x\to 0} \frac{\sin 3x}{x} = \lim_{x\to 0} 3\left(\frac{\sin 3x}{3x}\right) = 3\lim_{x\to 0} \frac{\sin 3x}{3x} = 3(1) = 3.$$

Example: Find $\displaystyle\lim_{x\to 0} \frac{1-\cos 2x}{\sin 3x}$.

Solution: Since $\displaystyle\frac{1-\cos 2x}{\sin 3x} = 2x\left(\frac{1-\cos 2x}{2x}\right)\frac{3x}{\sin 3x}\left(\frac{1}{3x}\right) = \frac{2}{3}\left(\frac{1-\cos 2x}{2x}\right)\left(\frac{3x}{\sin 3x}\right)$. Then

$$\lim_{x\to 0} \frac{1-\cos 2x}{\sin 3x} = \frac{2}{3}\left(\lim_{x\to 0}\left(\frac{1-\cos 2x}{2x}\right)\left(\lim_{x\to 0}\frac{3x}{\sin 3x}\right)\right) = \frac{2}{3}(0)(1) = 0.$$

Example: Find $\displaystyle\lim_{x\to \frac{\pi}{4}} \frac{\sin x}{\cos x}$.

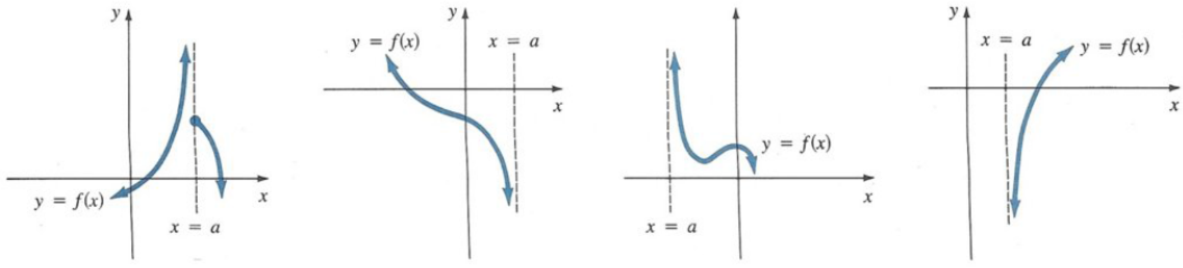
Solution: $\displaystyle\lim_{x\to \frac{\pi}{4}} \frac{\sin x}{\cos x} = \frac{\sin\left(\frac{\pi}{4}\right)}{\cos\left(\frac{\pi}{4}\right)} = 1.$

You Try It: Show that $\displaystyle\lim_{\theta\to 0} \frac{1-\cos\theta}{\theta} = 0.$

Limits at Infinity

It sometimes happen that as $x \to a$, $f(x)$ increases or decreases without bound. We write $\lim_{x\to a} f(x) = +\infty$ or $\lim_{x\to a} f(x) = -\infty$. We say that, $\lim_{x\to a} f(x) = +\infty$, if for each positive number M we can find a positive number δ (depending on M in general) such that $f(x) > M$ whenever $0 < |x-a| < \delta$.

Similarly, we say that $\lim_{x\to a} f(x) = -\infty$, if for each positive number M we can find a positive number δ (depending on M in general) such that $f(x) < -M$ whenever $0 < |x-a| < \delta$.



(a) $\lim_{x\to a^-} f(x) = \infty$ (b) $\lim_{x\to a^-} f(x) = -\infty$ (c) $\lim_{x\to a^+} f(x) = \infty$ (d) $\lim_{x\to a^+} f(x) = -\infty$

Note that $\lim_{x\to\infty} \frac{1}{x} = 0$ and $\lim_{x\to-\infty} \frac{1}{x} = 0$.

Limits at Infinity of a Rational Function

A rational function is a quotient of two polynomials, $f(x) = \dfrac{p_m(x)}{q_n(x)}$, where m and n are the degrees of the two polynomials.

1. **If** $m < n$, then $\lim_{x\to\infty} f(x) = 0$.

   Example: Find $\lim_{x\to\infty} \dfrac{x+1}{x^2+4}$.

   Solution: The degree of the numerator is one, the degree of the denominator is two. Therefore $\lim_{x\to\infty} \dfrac{x+1}{x^2+4} = 0$, since $\lim_{x\to\infty} \dfrac{\frac{1}{x}+\frac{1}{x^2}}{1+\frac{4}{x^2}} = \dfrac{0+0}{1+0} = 0$.

2. **If** $m > n$, then $\lim_{x\to\infty} f(x) = \pm\infty$. (sign depends on the polynomials $p_m(x)$ and $q_n(x)$, if they are of the same sign as x gets larger, the quotient is positive, if they are of opposite signs, the quotient is negative)

   Example: Find $\lim_{x\to\infty} \dfrac{x^3-2x^2+3x+4}{3x+5}$.

   Solution: $\lim_{x\to\infty} \dfrac{x^3-2x^2+3x+4}{3x+5} = \lim_{x\to\infty} \dfrac{1-\frac{2}{x}+\frac{3}{x^2}+\frac{4}{x^3}}{\frac{3}{x^2}+\frac{5}{x^3}} = \dfrac{1}{0} = \infty$.

3. **If** $m = n$, then $\lim_{x\to\infty} f(x) = \dfrac{a}{b}$, where a is the coefficient of x^m in the numerator and b is the coefficient of x^n in the denominator.

Example: Find $\lim_{x\to\infty} \frac{x^3-4x+1}{3x^3+2x+7}$.

Solution: $\lim_{x\to\infty} \frac{x^3-4x+1}{3x^3+2x+7} = \lim_{x\to\infty} \frac{1-\frac{4}{x^2}+\frac{1}{x^3}}{3+\frac{2}{x^2}+\frac{7}{x^3}} = \frac{1-0+0}{3+0+0} = \frac{1}{3}$.

You Try It: What is $\lim_{x\to\infty} \frac{a_0x^m+a_1x^{m-1}+\cdots+a_m}{b_0x^n+b_1x^{n-1}+\cdots+b_n}$, where $a_0, b_0 \neq 0$ and m and n are positive integers, when (a) $m > n$ (b) $m = n$ (c) $m < n$.

You Try It: Find $\lim_{x\to 4} \frac{x-4}{\sqrt{x}-2}$.

4.5 Continuity

A function $f(x)$ is **continuous** at a point $x = a$ if

1. $f(a)$ is defined.
2. $\lim_{x\to a} f(x)$ exists.
3. $\lim_{x\to a} f(x) = f(a)$.

Notice that, for $f(x)$ to be continuous at $x = a$, all three conditions must be satisfied. If at least one condition fails, f is said to have a discontinuity at $x = a$. For example, $f(x) = x^2 + 1$ is continuous at $x = 2$ since $\lim_{x\to 2} f(x) = 5 = f(2)$. The first condition above implies that a function can be continuous only at points of its domain. Thus, $f(x) = \sqrt{4-x^2}$ is not continuous at $x = 3$ because $f(3)$ is imaginary, i.e., not defined.

A function f is right-continuous (continuous from the right) at a point $x = a$ in its domain if $\lim_{x\to a^+} f(x) = f(a)$. It is left-continuous (continuous from the left) at $x = a$ if $\lim_{x\to a^-} f(x) = f(a)$. A function is continuous at an interior point $x = a$ of its domain if and only if it is both right-continuous and left-continuous at $x = a$.

Example: Determine whether $f(x) = x^2 + 1$ is continuous at $x = 1$.

Solution: $\lim_{x\to 1} x^2 + 1 = f(1) = 1^2 + 1 = 2$. Therefore $f(x) = x^2 + 1$ is continuous at $x = 1$.

Example: Determine whether $f(x) = \frac{|x|}{x}$ is continuous at $x = 0$.

Solution: Since $f(0)$ is not defined, $f(x)$ is not continuous at $x = 0$.

Example: Determine whether

$$f(x) = \begin{cases} \dfrac{|x|}{x}, & \text{if} \quad x \neq 0 \\ 0, & \text{if} \quad x = 0, \end{cases}$$

is continuous at $x = 0$.

Solution: $f(0)$ now defined. Then

Then $\lim_{x \to 0} f(x)$ must be considered in two steps,

$$\begin{aligned} \lim_{x \to 0^-} f(x) &= \lim_{x \to 0^-} (-1) = -1. \\ \lim_{x \to 0^+} f(x) &= \lim_{x \to 0^+} (1) = 1. \end{aligned}$$

Sine the limits are not the same, $\lim_{x \to 0} f(x)$ does not exist and $f(x)$ is not continuous at $x = 0$.

You Try It: Determine whether the function defined by

$$f(x) = \begin{cases} x^2, & \text{if} \quad x < 2 \\ 5, & \text{if} \quad x = 2 \\ -x + 6, & \text{if} \quad x > 2, \end{cases}$$

is continuous at the point $x = 2$.

A function $f(x)$ is **discontinuous** at $x = a$ if one or more of the conditions for continuity fails there.

Example: (a) $f(x) = \dfrac{1}{x-2}$ is discontinuous at $x = 2$, because $f(2)$ is not defined (has a zero denominator) and because $\lim_{x \to 2} f(x)$ does not exist (equals ∞). The function is, however, continuous everywhere except at $x = 2$, where it is said to have an *infinite discontinuity.*

(b) $f(x) = \dfrac{x^2 - 4}{x - 2}$ is discontinuous at $x = 2$ because $f(2)$ is not defined (both numerator and denominator are zero) and because $\lim_{x \to 2} f(x) = 4$. The discontinuity here is called *removable* since it may be removed by redefining the function as $f(x) = \dfrac{x^2 - 4}{x - 2}$ for $x \neq 2$ and $f(2) = 4$. (Note the discontinuity in (a) cannot be removed because the limit also does not exist.)

4.6 The $\varepsilon - \delta$ Definition of Continuity

$f(x)$ is continuous at $x = a$, if for any $\varepsilon > 0$, we can find $\delta > 0$, such that, $|f(x) - f(a)| < \varepsilon$ whenever $0 < |x - a| < \delta$.

Example: Prove that $f(x) = x^2$ is continuous at $x = 2$.

Solution: Must show that, given any $\varepsilon > 0$, we can find $\delta > 0$, such that $|f(x) - f(2)| = |x^2 - 4| < \varepsilon$ when $|x - 2| < \delta$.

Choose $\delta \leq 1$, so that $|x - 2| < 1$ or $1 < x < 3$ $(x \neq 2)$. Then $|x^2 - 4| = |(x-2)(x+2)| = |x-2||x+2| < \delta|x+2| < 5\delta$. Taking $\delta = \min\{1, \frac{\varepsilon}{5}\}$ whichever is smaller, then we have $|x^2 - 4| < \varepsilon$ whenever $|x - 2| < \delta$.

You Try It: (a) Prove that $f(x) = x$ is continuous at any point $x = x_0$.
(b) Prove that $f(x) = 2x^3 + x$ is continuous at any point $x = x_0$.

Theorems on Continuity

Theorem 1. If $f(x)$ and $g(x)$ are continuous at $x = a$, so are the functions $f(x) \pm g(x)$, $f(x)g(x)$ and $\dfrac{f(x)}{g(x)}$ if $g(x) \neq 0$.

Theorem 2. The following functions are continuous in every finite interval (a) all polynomials (b) $\sin x$ and $\cos x$ (c) $a^x, a > 0$.

Theorem 3. If $y = f(x)$ is continuous at $x = a$ and $z = g(y)$ is continuous at $y = b$ and if $b = f(a)$, then the function $z = g[f(x)]$ called a function of a function or composite function is continuous at $x = a$.

Briefly: A continuous function of a continuous function is continuous.

Theorem 4. If $f(x)$ is continuous in a closed interval, it is bounded in the interval.

You Try It: Suppose that $f(x) \leq g(x) \leq h(x)$ for all x in an open interval containing a and $\lim\limits_{x\to a} f(x) = \lim\limits_{x\to a} h(x) = L$. Then, show that $\lim\limits_{x\to a} g(x) = L$.

4.7 Tutorial Questions

1. Let
$$f(x) = \begin{cases} 3x - 1, & x < 0 \\ 0, & x = 0 \\ 2x + 5, & x > 0, \end{cases}$$
Evaluate (i) $\lim\limits_{x\to 2} f(x)$ (ii) $\lim\limits_{x\to -3} f(x)$ (iii) $\lim\limits_{x\to 0^+} f(x)$ (iv) $\lim\limits_{x\to 0^-} f(x)$ (v) $\lim\limits_{x\to 0} f(x)$.

2. If $f(x) = \dfrac{3x + |x|}{7x - 5|x|}$. Evaluate (i) $\lim\limits_{x\to\infty} f(x)$ (ii) $\lim\limits_{x\to -\infty} f(x)$ (iii) $\lim\limits_{x\to 0^+} f(x)$ (iv) $\lim\limits_{x\to 0^-} f(x)$ (v) $\lim\limits_{x\to 0} f(x)$

3. Use the theorems on limits to evaluate each of the following limits.
(i) $\lim\limits_{x\to 0}\dfrac{x^3-8}{x-2}$ (ii) $\lim\limits_{x\to 0}\dfrac{\sqrt{3+x}-\sqrt{3}}{x}$ (iii) $\lim\limits_{x\to\infty}\dfrac{8x^2+4}{2x^2-x+2}$ (iv) $\lim\limits_{x\to 2}(4x^2-x+5)$
(v) $\lim\limits_{x\to 0}\dfrac{\sqrt{x+4}-2}{x}$ (vi) $\lim\limits_{x\to\infty}\dfrac{a_0+a_1x+\cdots+a_mx^m}{b_0+b_1x+\cdots+b_nx^n}$ (vii) $\lim\limits_{x\to\infty}\dfrac{(3x-1)(2x+3)}{(5x-3)(4x+5)}$
(viii) $\lim\limits_{x\to 0}\dfrac{\sin 3x}{x}$ (ix) $\lim\limits_{x\to 0}\dfrac{6x-\sin 2x}{2x+3\sin 4x}$ (x) $\lim\limits_{x\to 0}\dfrac{\sin x}{\sin 2x}$ (xi) $\lim\limits_{x\to 4}\dfrac{2x-8\sqrt{x}+8}{\sqrt{x}-2}$.

4. For $f(x)=5x-6$, find a $\delta>0$ such that whenever $0<|x-4|<\delta$, then $|f(x)-14|<\varepsilon$ when (i) $\varepsilon=\frac{1}{2}$ (ii) $\varepsilon=0.001$.

5. Verify (i) $\lim\limits_{x\to 2}\dfrac{x^2-4}{x-2}=4$ (ii) $\lim\limits_{x\to 0}x^2\cos\left(\dfrac{1}{x}\right)=0$.

6. Give the points of discontinuity of each of the following functions.
(i) $f(x)=\dfrac{x}{(x-2)(x-4)}$ (ii) $f(x)=x^2\sin\left(\frac{1}{x}\right), f(0)=0$
(iii) $f(x)=\sqrt{(x-3)(6-x)}, 3\le x\le 6$ (iv) $f(x)=\dfrac{1}{1+2\sin x}$.

7. For what values of x in the domain of definition is each of the following functions continuous?
(i) $f(x)=\dfrac{x}{x^2-1}$ (ii) $f(x)=\dfrac{1+\cos x}{3+\sin x}$ (iii) $f(x)=10^{-\frac{1}{(x-3)^2}}$
(iv) $f(x)=\dfrac{x-|x|}{x}$.

8. Given that the function $f:\mathbb{R}\to\mathbb{R}$ be defined by

$$f(x)=\begin{cases} x^2-4, & x\ge 2\\ 2ax+b, & 0<x<2\\ e^x, & x\le 0,\end{cases}$$

is continuous at all points in $\mathbb{R}$. Find the values of a and b.

9. Find the values of a and b such that the function f will be continuous on $\mathbb{R}$

$$f(x)=\begin{cases} ax-3, & x<2\\ 3-x-2x^2, & 2\le x\le 4\\ bx+7, & x>4.\end{cases}$$

10. Determine all values of the the constant A so that the following is continuous for all values of x,

$$f(x)=\begin{cases} A^2x-A, & x\ge 3\\ 4, & x<3.\end{cases}$$

11. Determine all the values of the constants A and B so that the following is continuous for all values of x,

$$f(x)=\begin{cases} Ax-B, & x\le -1\\ 2x^2+3Ax+B, & -1<x\le 1\\ 4, & x>1.\end{cases}$$

12. The function $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(x) = \begin{cases} \sin x, & x \leq 0 \\ ax + b, & 0 < x \leq 3 \\ x^3, & x > 3. \end{cases}$$

Find all the values of a and b if f is continuous on $\mathbb{R}$.

13. Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} x^2, & x \geq 2 \\ ax + b, & 1 < x < 2 \\ 2 - x, & x \leq 1, \end{cases}$$

where a and b are constants. If f is continuous on $\mathbb{R}$, find the values a and b.